

Trajectory-Adaptive Stopping Rules for SGD under Strong Convexity

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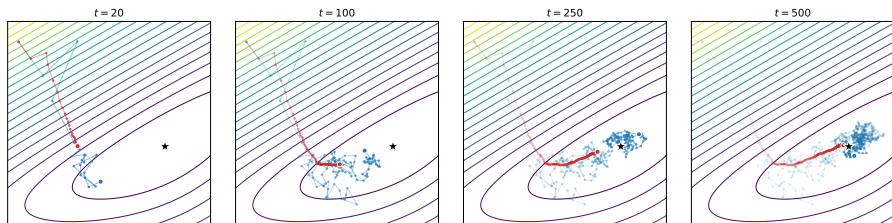
Beyond Optimal Rates in Stochastic Optimization: Trajectory-Adaptive Stopping Rules

Liviu Aolaritei, Lucas Lévy, Francis Bach, Michael I. Jordan

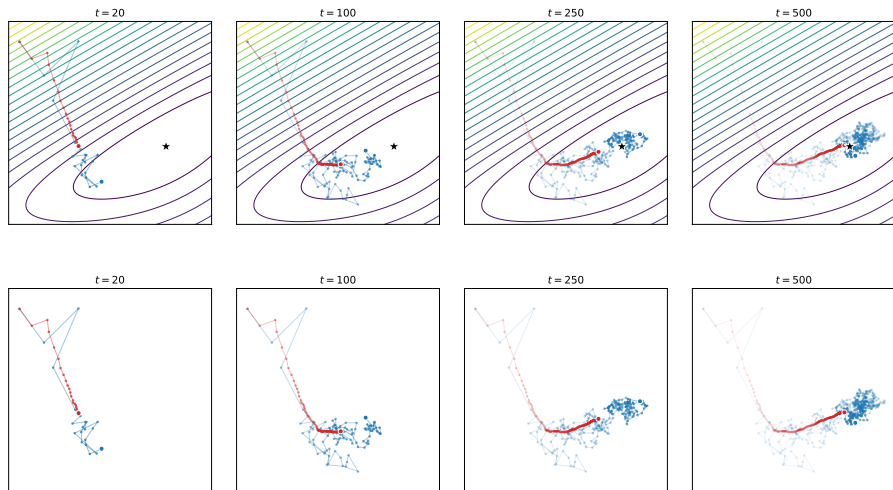
Under Review in Mathematical Programming. *arXiv preprint arXiv:2608.25551.*



When to stop SGD ?



When to stop SGD ?



- ▶ Too early $\implies$ missing a better solution
- ▶ Too late $\implies$ wasting computations

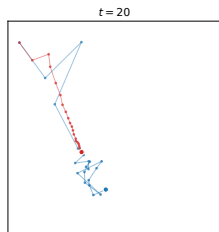
Fixed-horizon theory

Proposition [Rakhlin et al., 2012]: Let f be μ -strongly convex. Assume that $\|g_t\| \leq G$ a.s. and that we use stepsizes $\eta_t = 1/\mu t$. Then, for a fixed T ,

$$\mathbb{P}\left[\|x_T - x^*\|^2 \leq \frac{(624 \ln(\ln(T)/\delta) + 1)G^2}{\mu^2 T}\right] \geq 1 - \delta$$

- ▶ Identified optimal rates in T , influence of strong convexity, choice of step-size schedules, influence of averaging, ...
- ▶ But practitioners often face the inverse question

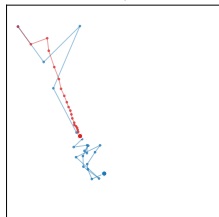
Continuous monitoring



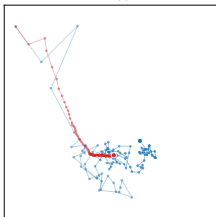
- ▶ Should I stop ?
- ▶ Should I continue ?
- ▶ Should I restart with new parameters ?

Continuous monitoring

$t = 20$



$t = 100$



- ▶ Should I stop ?
- ▶ Should I continue ?
- ▶ Should I restart with new parameters ?

- ▶ Only access to proxies (e.g. validation loss)
- ▶ Fixed-horizon high-probability guarantees no longer hold under optional stopping:

$$\left\{ \forall t, \mathbb{P}(Z_t \leq U_t) \geq 1 - \delta \right\} \not\Rightarrow \mathbb{P}(Z_\tau \leq U_\tau) \geq 1 - \delta, \quad \tau \text{ stopping time}$$

Stopping based on worst-case bounds

Inverting a fixed-time high-probability bound

$$\mathbb{P}[Z_t \leq U_t(\delta)] \geq 1 - \delta \quad \text{with } U_t(\delta) \text{ deterministic}$$

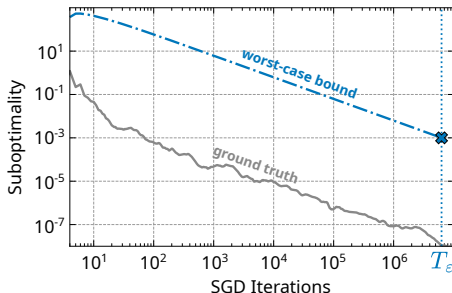
For $\varepsilon > 0$, stop at deterministic $T_\varepsilon := \{t : U_t(\delta) \leq \varepsilon\}$.

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Overly conservative: stops $\sim 10^5$ times too late

What a stopping certificate needs

Fixed-horizon guarantees

valid, but conservative

Heuristic early stopping

adaptive, but uncertified

We look for a certificate that is:

- ▶ **observable**: computed from the trajectory, not from x^* or f^*
- ▶ **time-uniform**: valid at any data-dependent stopping time
- ▶ **adaptive**: tight on the realized run, not the worst case

Can SGD certify, from its own realized trajectory, when it should stop?

Approach: treat the SGD run as a sequential experiment.

Previous work

- ▶ **Anytime-valid confidence sequences** (Ramdas et al. 2023, Howard et al. 2021):

$$\mathbb{P}[\forall t, Z_t \leq U_t(\delta)] \geq 1 - \delta$$

- ▶ Stays valid under optional stopping
 - ▶ U_t may depend on observed trajectory up to time t .
- ▶ Aolaritei and Jordan (2025): certificates in general convex case
- ▶ **This work:** strongly convex case, matching $\tilde{O}(1/t)$ rate requires new tools

Problem formulation

- ▶ μ -strongly convex function f with minimizer $x^\star$
- ▶ At each time t , update

$$x_{t+1} = x_t - \eta_t g_t \tag{1}$$

for step size $\eta_t \geq 0$ and stochastic gradient g_t .

- ▶ Stochastic gradients satisfy $\|g_t\| \leq G$ and $\mathbb{E}_{t-1}[g_t] = f'(x_t)$
- ▶ **Goal:** Establish an observable AVUCS on $Z_t := \|x_t - x^\star\|^2$.

Classical Analysis of SGD

Following Nemirovski et al. (2009), for all t ,

$$Z_{t+1} \leq (1 - 2\mu\eta_t)Z_t + \eta_t^2\|g_t\|^2 + 2\eta_t\langle x^* - x_t, g_t - f'(x_t) \rangle \quad (2)$$

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Then, defining $A_t := \prod_{s=t_0}^t (1 - 2\mu\eta_s)^{-1}$, we unroll the recursion

$$Z_{t+1} \leq \frac{1}{A_t} \left(R_0 + \underbrace{\sum_{s=t_0}^t A_s \eta_s^2 \|g_s\|^2}_{\text{Weighted sum of gradients norms (observable)}} + \underbrace{\sum_{s=t_0}^t 2A_s \eta_s \langle x^* - x_s, g_s - f'(x_s) \rangle}_{\text{Noise martingale}} \right)$$

Classical Analysis of SGD

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► **In-expectation** bound: $\mathbb{E}[\text{noise martingale}] = 0$ so

$$\mathbb{E}[Z_{t+1}] \leq \frac{1}{A_t} (R_0 + G^2 \sum_{s=t_0}^t A_s \eta_s^2) \asymp \frac{G^2}{\mu^2 t} \quad \text{for step-sizes } \eta_t = 1/\mu t$$

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- **High-probability** bound (Rakhlin et al., 2012): Applying Freedman's inequality,

$$\mathbb{P}\left[Z_t \leq \frac{(624 \ln(\ln(t)/\delta) + 1)G^2}{\mu^2 t}\right] \geq 1 - \delta$$

Anytime-valid Hoeffding

Assume X_t is v_t -**subgaussian** i.e.,

$$\mathbb{E}_{t-1}[\exp(\lambda(X_t - \mathbb{E}_{t-1}X_t) - \frac{1}{2}\lambda^2 v_t)] \leq 1$$

- By **independence**,

$$\mathbb{E}[\exp(\lambda \bar{S}_t - \frac{1}{2}\lambda^2 V_t)] \leq 1$$

with $\bar{S}_t := \sum_{s=1}^t (X_s - \mathbb{E}_{s-1}X_s)$ and

$$V_t := \sum_{s=1}^t v_s$$

- **Markov's inequality:**

$$\mathbb{P}[\exp(\lambda \bar{S}_t - \frac{1}{2}\lambda^2 V_t) \geq \frac{1}{\delta}] \leq \delta$$

- **Optimize λ :**

$$\mathbb{P}[\bar{S}_t \geq \sqrt{2V_t \ln(1/\delta)}] \leq \delta$$

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- ▶ The process
 $M_t^\lambda := \exp(\lambda \bar{S}_t - \frac{1}{2}\lambda^2 V_t)$ is a
non-negative supermartingale
- ▶ **Ville's inequality**:
 $\mathbb{P}[\exists t \geq 1, M_t^\lambda \geq \frac{1}{\delta}] \leq \delta$
- ▶ **Stitch** for several λ :
 $\mathbb{P}[\exists t \geq 1, \bar{S}_t \geq \mathfrak{H}_\delta(V_t)] \leq \delta$
with $\mathfrak{H}_\delta(v) \approx \sqrt{v(\ln(1/\delta) + \ln \ln v)}$

Hoeffding-based confidence sequence

With probability $1 - \delta$, simultaneously for all t ,

$$Z_{t+1} \leq \frac{1}{A_t} \left(R_0 + \sum_{s=t_0}^t A_s \eta_s^2 \|g_s\|^2 + \mathfrak{H}_\delta \left(4G^2 \sum_{s=t_0}^t A_s^2 \eta_s^2 Z_s \right) \right)$$

We obtain the recursively-defined confidence sequence:

$$\hat{U}_{t+1}(\delta) := \frac{1}{A_t} \left(R_0 + \sum_{s=t_0}^t A_s \eta_s^2 \|g_s\|^2 + \mathfrak{H}_\delta \left(4G^2 \sum_{s=t_0}^t A_s^2 \eta_s^2 \hat{U}_s(\delta) \right) \right)$$

We can show that for classical stepsizes,

$$\hat{U}_{t+1}(\delta) \asymp \frac{G^2}{\mu^2} \frac{\ln(\ln(t)/\delta)}{t}$$

Empirical Bernstein confidence sequence

- ▶ Hoeffding's inequality controlled the deviation of X_t from its mean by a deterministic worst-case quantity v :

$$\mathbb{E}[\exp(\lambda(X - \mathbb{E}X) - \frac{1}{2}\lambda^2 v)] \leq 1$$

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- ▶ Bernstein's inequality controls the deviation by the variance
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 - ▶ Requires a non-trivial bound on the variance of X_t

Empirical Bernstein confidence sequence

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- ▶ Bernstein's inequality controls the deviation by the variance
 - ▶ Asymptotically tighter than Hoeffding's
 - ▶ Requires a non-trivial bound on the variance of X_t
- ▶ Empirical Bernstein-type inequalities ([Audibert et al. 2007](#)) control the deviation by an online estimator of the variance

Howard's inequality ([Howard et al. 2021](#)): If $X \geq -1$ a.s., then for all $\lambda \in [0, 1)$,

$$\mathbb{E}[\exp(\lambda(X - \mathbb{E}X) - \psi_E(\lambda)X^2)] \leq 1$$

with $\psi_E(\lambda) = -\ln(1 - \lambda) - \lambda$.

Empirical Bernstein confidence sequence

Theorem (Informal). Let $\{X_t\}_t$ be a real process and $\{c_t\}_t$ be predictable such that $|X_t| \leq c_t$. Define

$$\bar{S}_t := \sum_{s=1}^t X_s - \mathbb{E}_{s-1}[X_s], \quad V_t := \sum_{s=1}^t X_s^2, \quad C_t := \max_{s \leq t} c_s$$

Then, for confidence level $1 - \delta$,

$$\mathbb{P}[\exists t \geq 1, \bar{S}_t \geq \mathfrak{B}_\delta(V_t, C_t)] \leq \delta$$

with $\mathfrak{B}_\delta(v, c) \asymp \sqrt{v (\ln \ln v + \ln \frac{1}{\delta})} + c (\ln \ln c + \ln \frac{1}{\delta})$.

Empirical Bernstein confidence sequence

Plugging this new inequality into our previous analysis, we get the CS:

$$\hat{U}_{t+1}^{\text{EB}}(\delta) := \frac{1}{A_t} \left(R_0 + \sum_{s=t_0}^t A_s \eta_s^2 \|g_s\|^2 + \mathfrak{B}_\delta(\hat{V}_t, \hat{C}_t) \right) \quad \text{with}$$

$$\hat{V}_t := 4 \sum_{s=t_0}^t A_s^2 \eta_s^2 \|g_s\|^2 \hat{U}_s^{\text{EB}}(\delta), \quad \text{and} \quad \hat{C}_t := 2G \max_{s \leq t} A_s \eta_s \sqrt{\hat{U}_s^{\text{EB}}(\delta)}$$

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If $\frac{1}{t} \sum_{s=t_0}^t \|g_s\|^2 \rightarrow \mathbf{g}$, it can be shown that

$$\hat{U}_{t+1}^{\text{EB}}(\delta) \asymp \frac{\mathbf{g}^2 \ln(\ln(t)/\delta)}{\mu^2 t}$$

Improvements under minibatching

Minibatch SGD:

$$x_{t+1} = x_t - \eta_t \frac{1}{b} \sum_{i=1}^b g_t^i$$

Assumptions:

1. Independence of $g_t^1, \dots, g_t^b$ conditionally on $\mathcal{F}_{t-1}$
2. Average unbiasedness: $\mathbb{E}_{t-1}[\frac{1}{b} \sum_{i=1}^b g_t^i] = f'(x_t)$
3. Uniform boundedness: $\|g_t^i\| \leq G$ a.s. for all t, i

Improvements under minibatching

Theorem (Informal). Let $\{X_t^{(b)}\}_t$ be b real integrable processes and $\{c_t\}_t$ be predictable such that

1. $X_t^{(1)}, \dots, X_t^{(b)}$ are independent conditionally on $\mathcal{F}_{t-1}$.
2. $|X_t^{(i)}| \leq c_t$ a.s. for all i .

Define

$$M_t := \sum_{s=1}^t \sum_{i=1}^b X_s^{(i)} - \mathbb{E}_{s-1}[X_s^{(i)}], \quad W_t := \sum_{s=1}^t \sum_{i=1}^b (X_s^{(i)})^2, \quad C_t := \max_{s \leq t} c_s$$

Then, for confidence level $1 - \delta$,

$$\mathbb{P}[\exists t \geq 1, M_t \geq \mathfrak{B}_\delta(W_t, C_t)] \leq \delta$$

Improvements under minibatching

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Application to Confidence Sequences for Z_t :

- ▶ Dependence on $\lambda_{\max}(\sum_{i=1}^b g_t^i \otimes g_t^i)$ rather than $\|\bar{g}_t\|^2$
- ▶ The contribution of G is divided by b

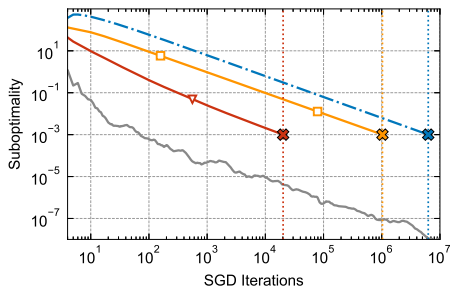
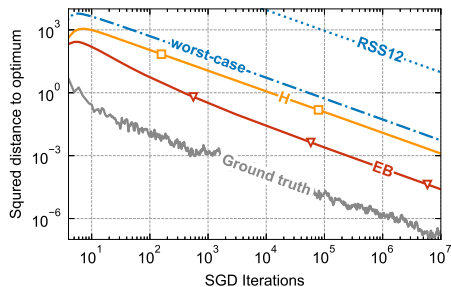
Experimental setup

SVM primal problem:

$$f(x) = \mathbb{E}_z[\max\{0, 1 - z^\top x\} + \frac{\mu}{2}\|x\|^2]$$

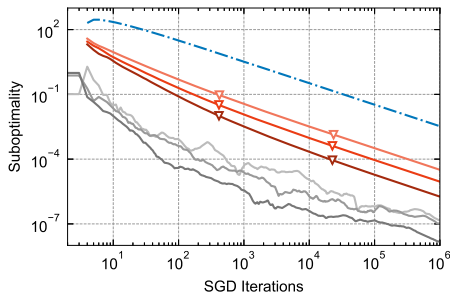
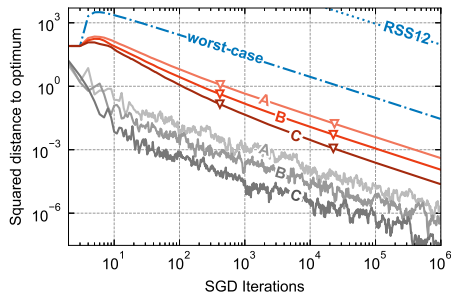
- ▶ $\mu = 0.1$, $1 - \alpha = 99\%$, $b = 256$
- ▶ Optimal solution computed beforehand with LIBLINEAR
- ▶ Comparison for both $Z_t = \|x_t - x^*\|^2$ and $f(\bar{x}_t) - f(x^*)$.
- ▶ Visualisations on one single run

Experimental result on real-world data



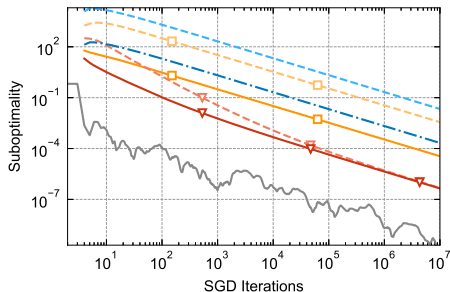
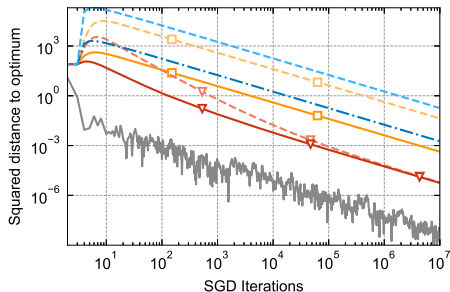
On the right, crosses mark the 1st time each bound falls below $\epsilon = 10^{-3}$ threshold.

Adaptivity across problem instances



SGD on 3 datasets generated with `make_classification`, with increasing classes separability from A to C.

Robustness to parameter misspecification



- ▶ Solid and dash-dotted lines: correct value of G .
- ▶ Dashed lines in lighter shades: G over-estimated by a factor 10.

Conclusion

- ▶ We have established **anytime-valid**, **observable** and **adaptive** confidence sequences for SGD
- ▶ We have developed a **novel empirical-Bernstein-based confidence sequence** for processes with predictable range.

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- ▶ There is still a gap between our CS and the ground truth. Is it solvable using anytime-valid inference tools ?
- ▶ Other settings than strong convexity ?

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paper & slides



Thank you for your attention

References

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